

GAYATHRI KARAGODA PATHIRANAGE

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PROFILE

Data Analyst with an MSc in Statistics and Data Analytics, experienced in SQL, Python, and data visualization. Skilled in analyzing large datasets, building interactive dashboards, and applying statistical and machine learning techniques to generate actionable insights. Strong communicator with the ability to support data-driven decision-making.

TECHNICAL SKILLS

Programming & Databases: Python (Pandas, NumPy, Matplotlib), SQL (MySQL, SQL Server), R

Data Visualization: Tableau, Power BI, Excel (Pivot Tables, VLOOKUP)

Statistical & Machine Learning Techniques: Regression (Linear, Logistic), Bayesian Inference, Predictive Modelling, Time Series (ARIMA)

Tools & Techniques: EDA, Data Cleaning & Preprocessing, Dashboard Development, Data Storytelling

PROJECTS

UK Rail Performance & Delay Analysis Dashboard

- Analyzed 30,000+ train journey records using SQL and Python
- Built an interactive Tableau dashboard for travel demand, delays, and revenue
- Identified peak delay periods and key causes such as weather and signal failures
- Impact:** Provided actionable insights to improve scheduling and operational efficiency

England & Wales Property Price Analysis

- Performed SQL-based analysis on property transaction datasets
- Developed Tableau dashboard for regional price trends and comparisons
- Analyzed property types, tenure, and long-term price growth
- Impact:** Revealed regional disparities and key market trends

Apple Stock Price Analysis & Forecasting

- Built predictive models using ARIMA and regression in Python
- Compared model performance using MSE
- Developed Power BI dashboard to visualize forecasts
- Impact:** Improved short-term forecasting accuracy using lag-based models

Energy Consumption Prediction (Bayesian Logistic Regression)

- Developed Bayesian model using MCMC in R
 - Predicted probability of high energy consumption
 - Evaluated model performance using classification metrics
 - Impact:** Enabled prediction under uncertainty using probabilistic modelling
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EDUCATION

MSc Statistics with Data Analytics - Brunel University London, UK - 2023

BSc Mathematics, Mathematical Modelling & Statistics - Wayamba University of Sri Lanka - 2017